

SWIFT MOTORS – VEHICLE MANUFACTURING & QUALITY ANALYTICS

IBM Cognos Analytics Case Study Project

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Tools: IBM Cognos Analytics, Data Visualization, Dashboarding

Domain: Manufacturing Analytics & Business Intelligence

Executive Summary

This project analyzes manufacturing operations and quality metrics for Swift Motors using IBM Cognos Analytics.

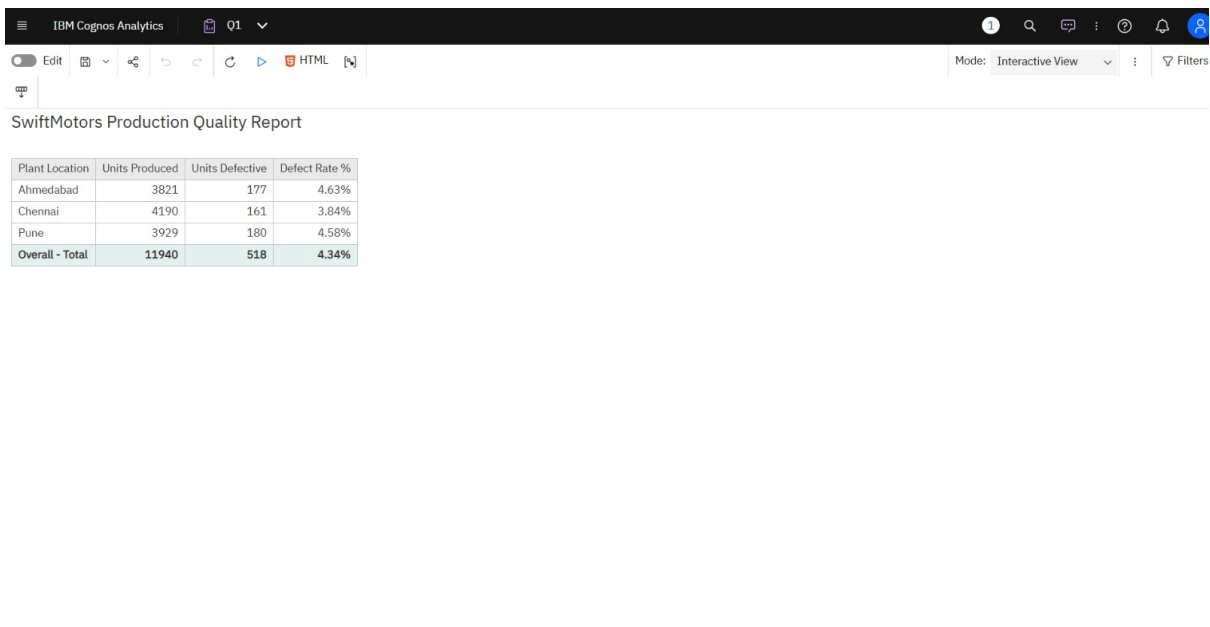
Key Areas:

- Production Quality Analysis
- Downtime Monitoring
- Defect Analysis
- Supplier Performance Analysis
- KPI Dashboard Development
- Executive Storytelling

Skills Demonstrated

- Data Analysis
- Dashboard Development
- KPI Reporting
- Data Visualization
- Business Intelligence
- Manufacturing Analytics
- Exploratory Data Analysis
- Executive Reporting
- IBM Cognos Analytics

1. Production Quality Report



IBM Cognos Analytics Q1

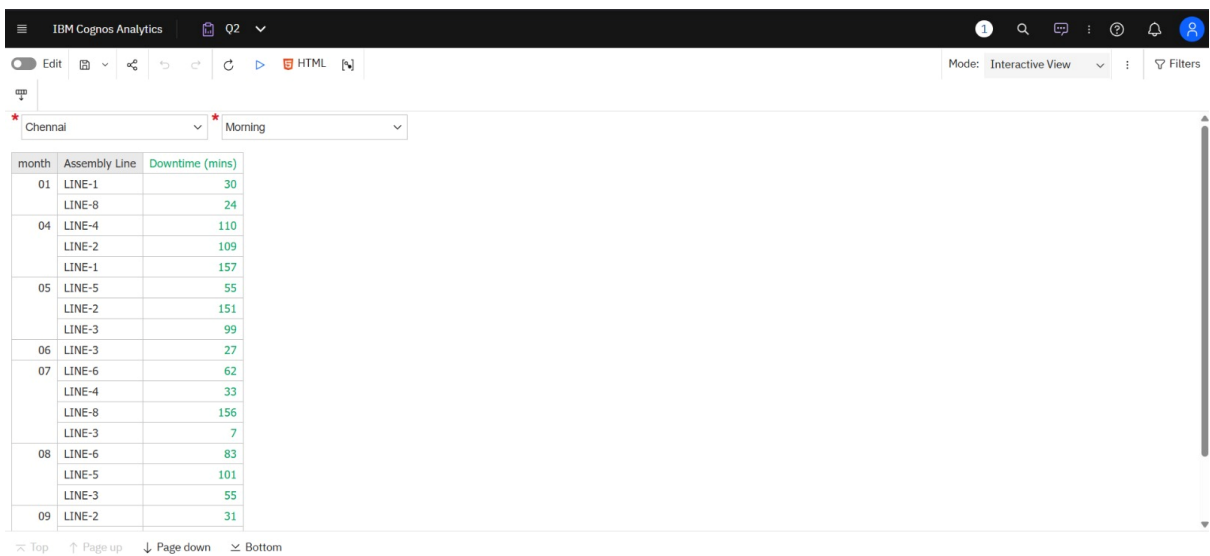
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SwiftMotors Production Quality Report

Plant Location	Units Produced	Units Defective	Defect Rate %
Ahmedabad	3821	177	4.63%
Chennai	4190	161	3.84%
Pune	3929	180	4.58%
Overall - Total	11940	518	4.34%

Production Quality Report displaying units produced, defective units, and defect rate percentages across manufacturing plants, enabling quality performance monitoring and defect tracking

2. Downtime Analysis Report



IBM Cognos Analytics Q2

Mode: Interactive View Filters

* Chennai * Morning

month	Assembly Line	Downtime (mins)
01	LINE-1	30
	LINE-8	24
04	LINE-4	110
	LINE-2	109
	LINE-1	157
05	LINE-5	55
	LINE-2	151
	LINE-3	99
06	LINE-3	27
07	LINE-6	62
	LINE-4	33
	LINE-8	156
	LINE-3	7
08	LINE-6	83
	LINE-5	101
	LINE-3	55
09	LINE-2	31

⌂ Top ↑ Page up ↓ Page down ⌂ Bottom

IBM Cognos Analytics

Q2

1

Q

Edit

HTML

Mode: Interactive View

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Prompt

Provide values for the report you are about to run.

plant

* plant

shift

* shift

OK

Cancel

IBM Cognos Analytics

Q2

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Mode: Interactive View

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* Chennai

* Morning

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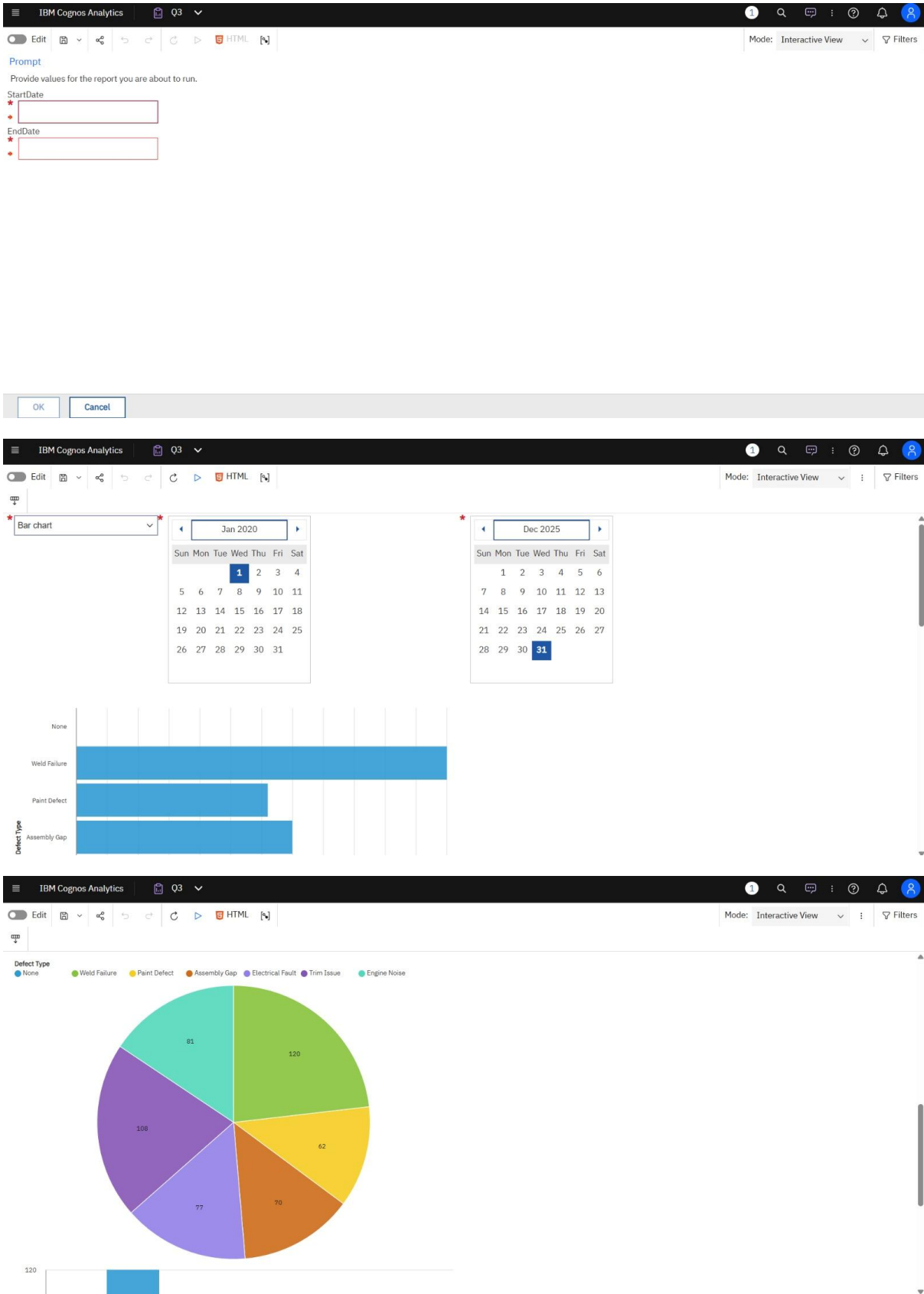
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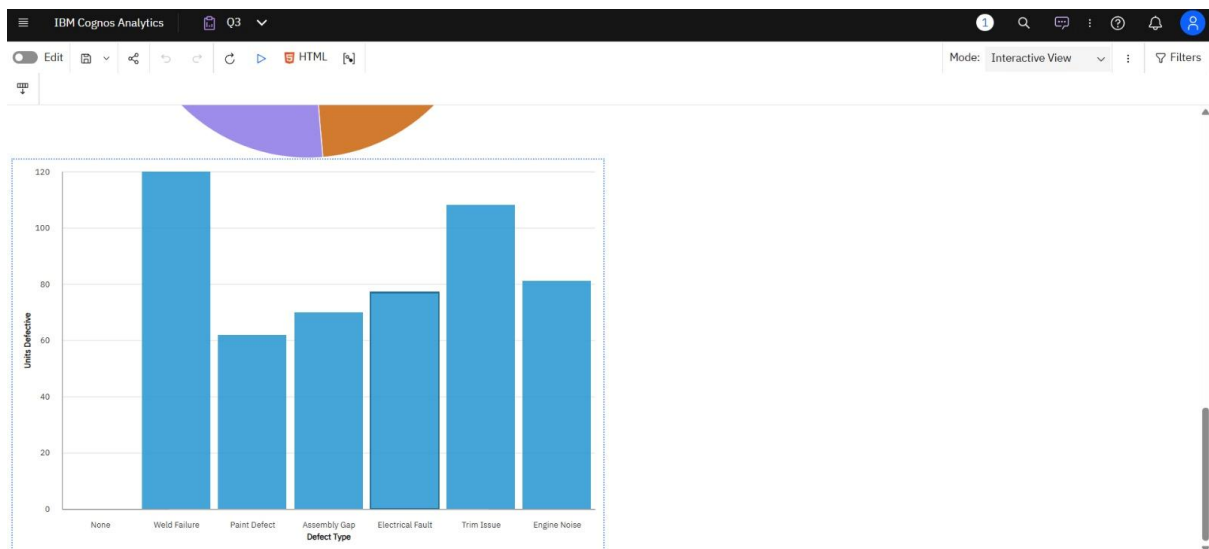
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Downtime Analysis Report highlighting assembly-line downtime trends and production losses to identify operational bottlenecks and improve manufacturing efficiency.

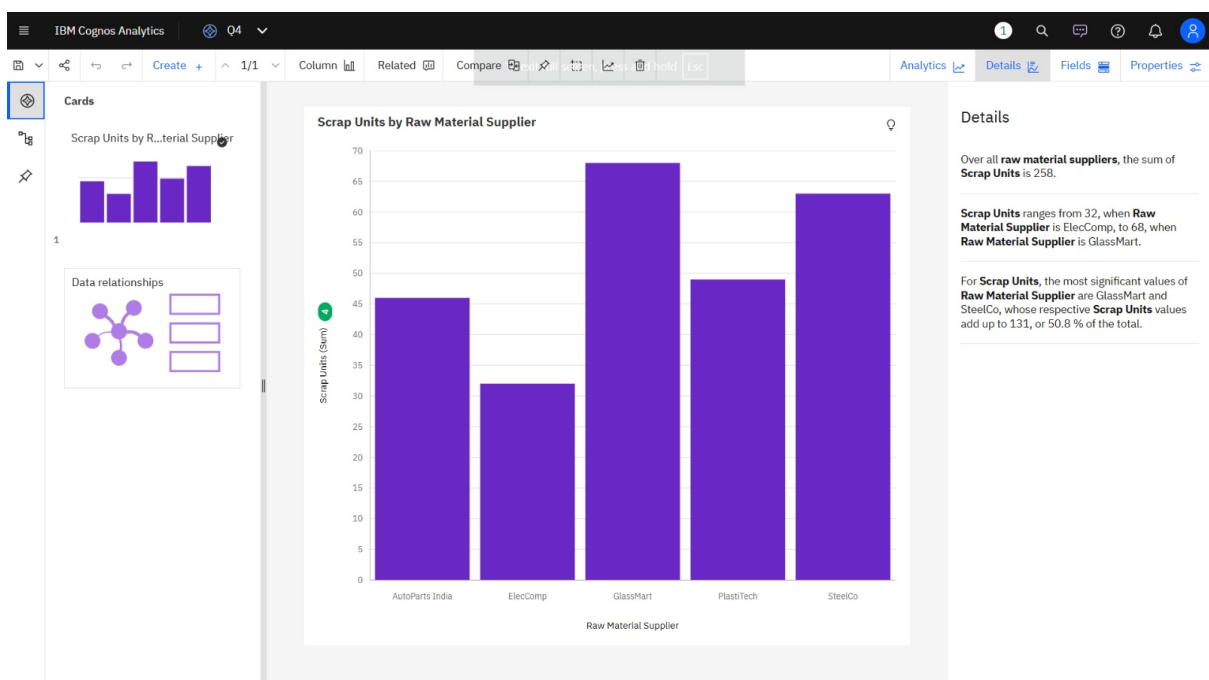
3. Defect Type Distribution





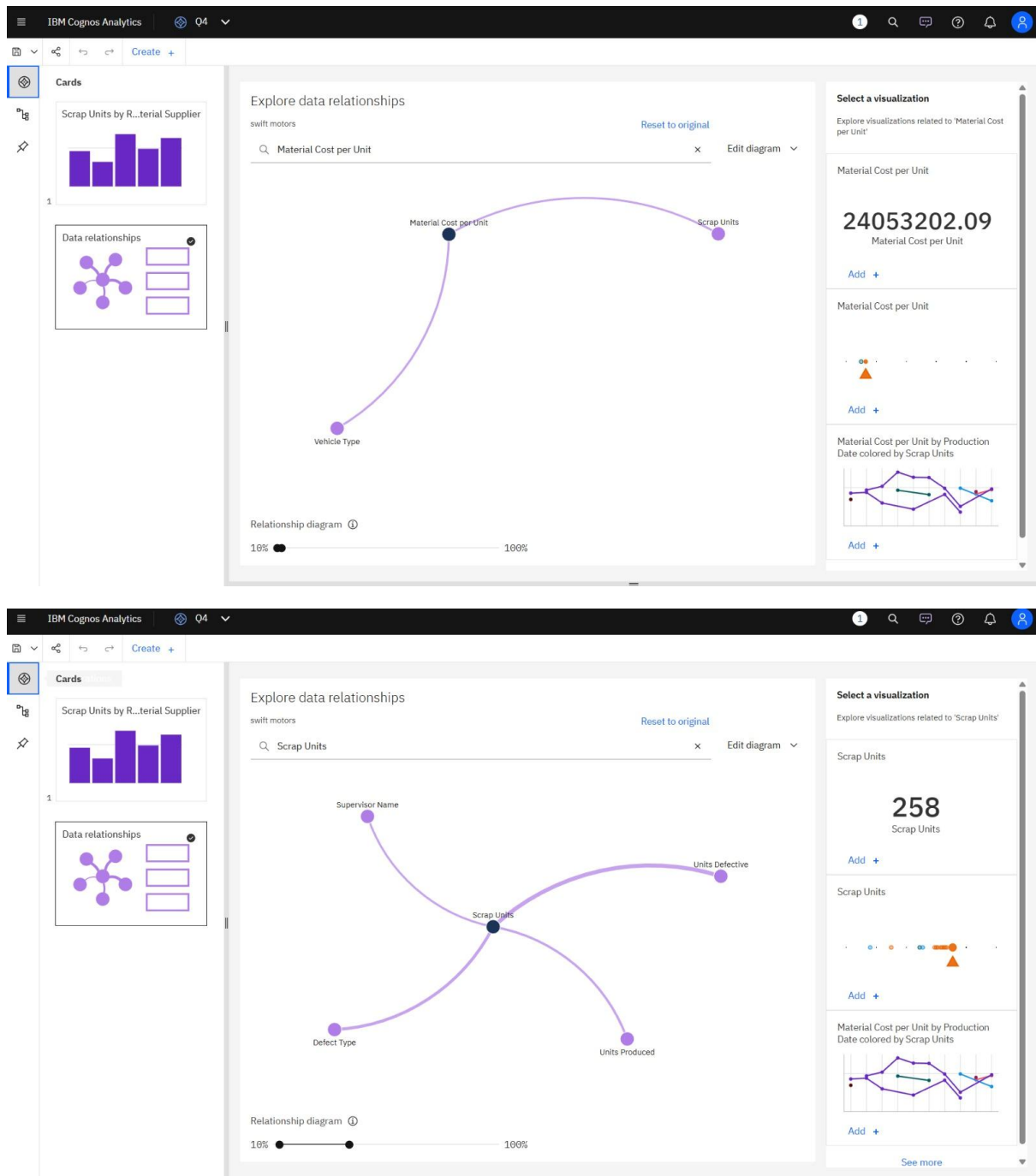
Interactive visualization of defect type distribution across manufacturing operations, supporting root-cause analysis and quality improvement initiatives.

4. Exploratory Analytics – Supplier Analysis



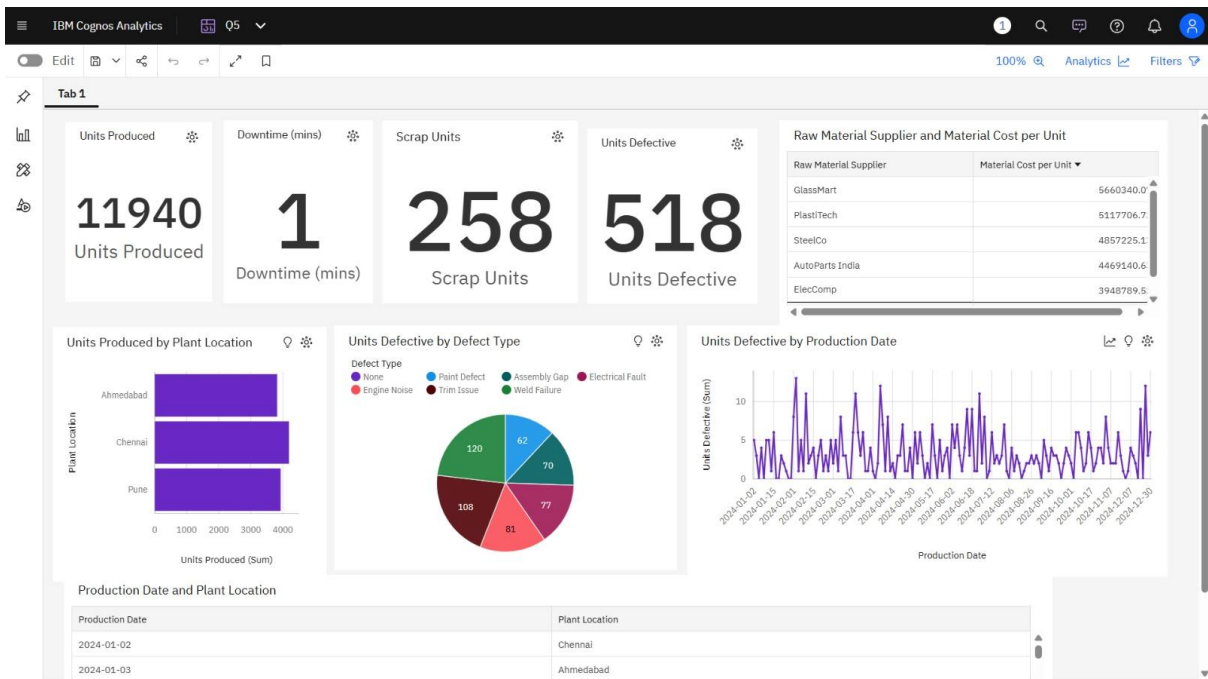
Exploratory analytics examining relationships between raw material suppliers and scrap units to uncover factors influencing manufacturing quality and operational costs.

5. Exploratory Analytics – Shift & Defect Analysis



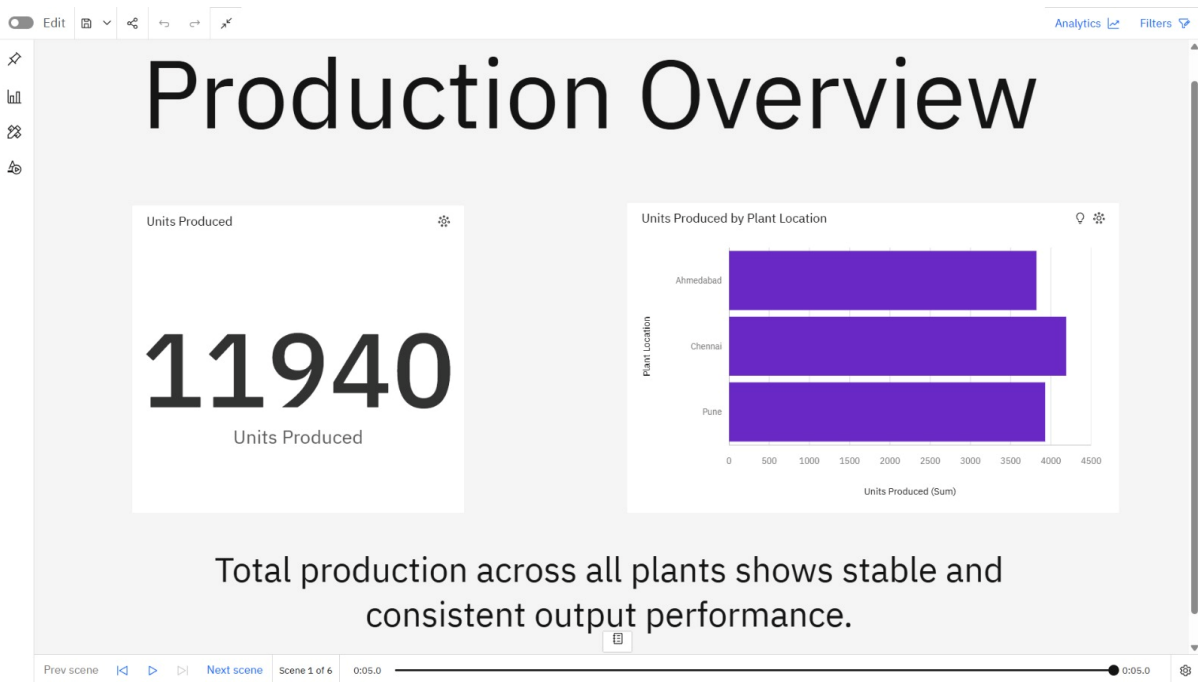
Relationship analysis between shift schedules, assembly-line performance, and defect rates to identify patterns affecting production quality.

6. Manufacturing Operations Dashboard



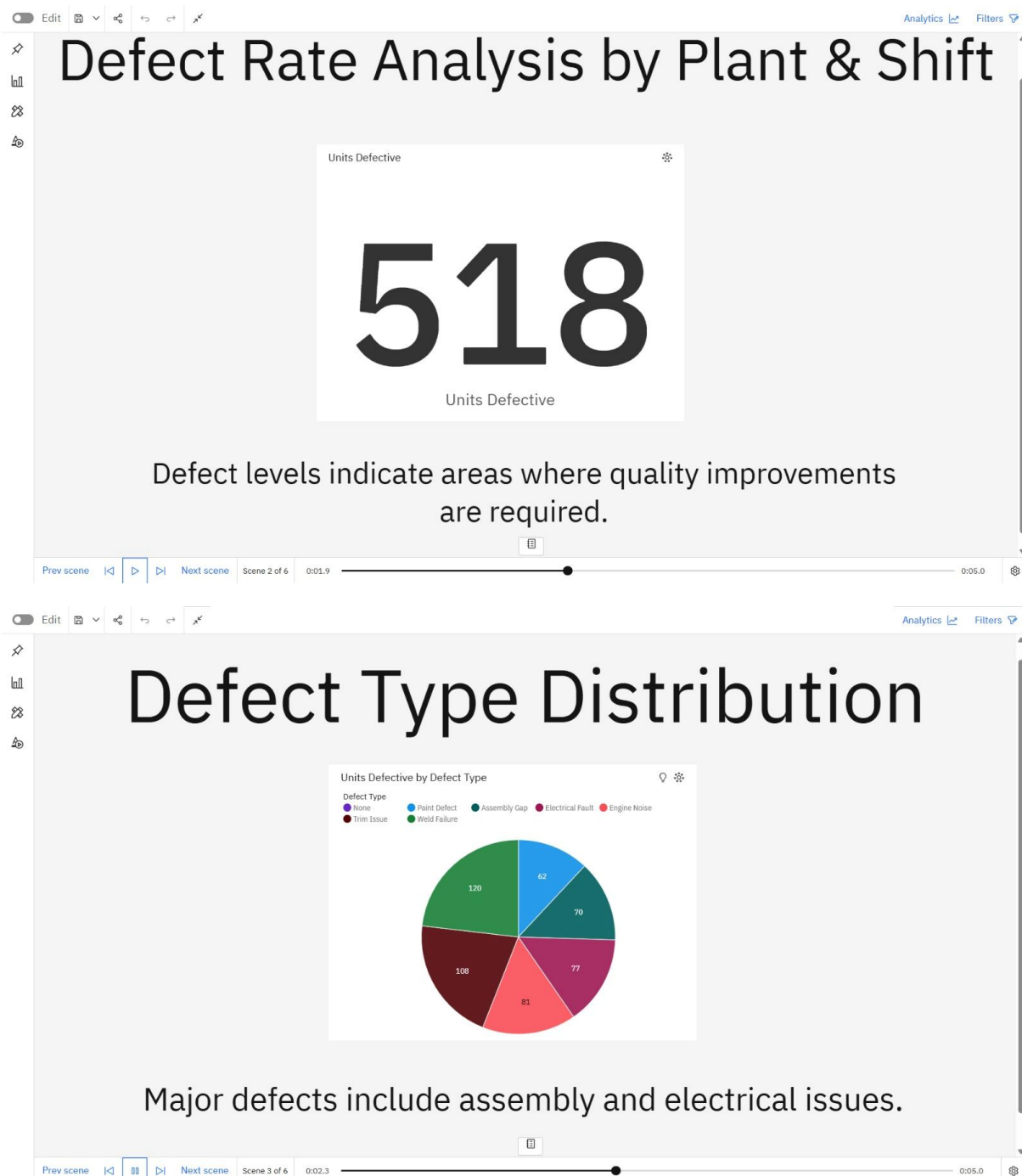
Interactive manufacturing dashboard featuring KPI metrics, production trends, defect analysis, supplier performance, and operational insights for data-driven decision-making.

7. Executive Story – Production Overview



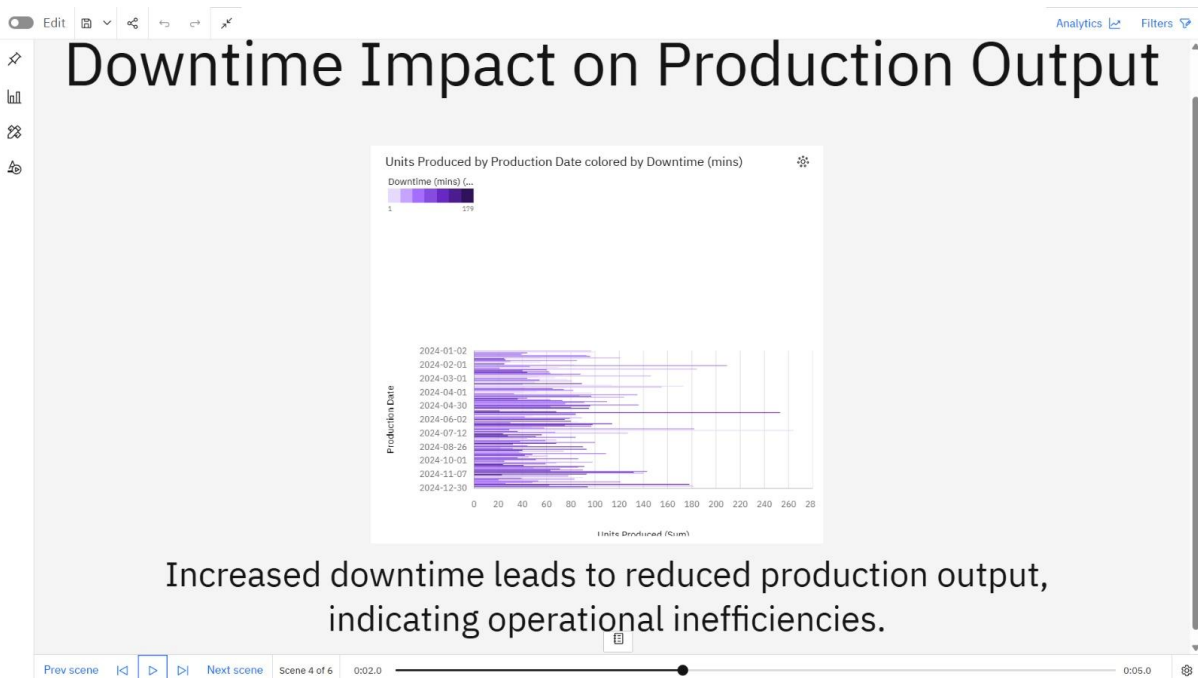
Executive-level production overview summarizing manufacturing output across plants and providing a high-level view of operational performance.

8. Executive Story – Defect Rate Analysis



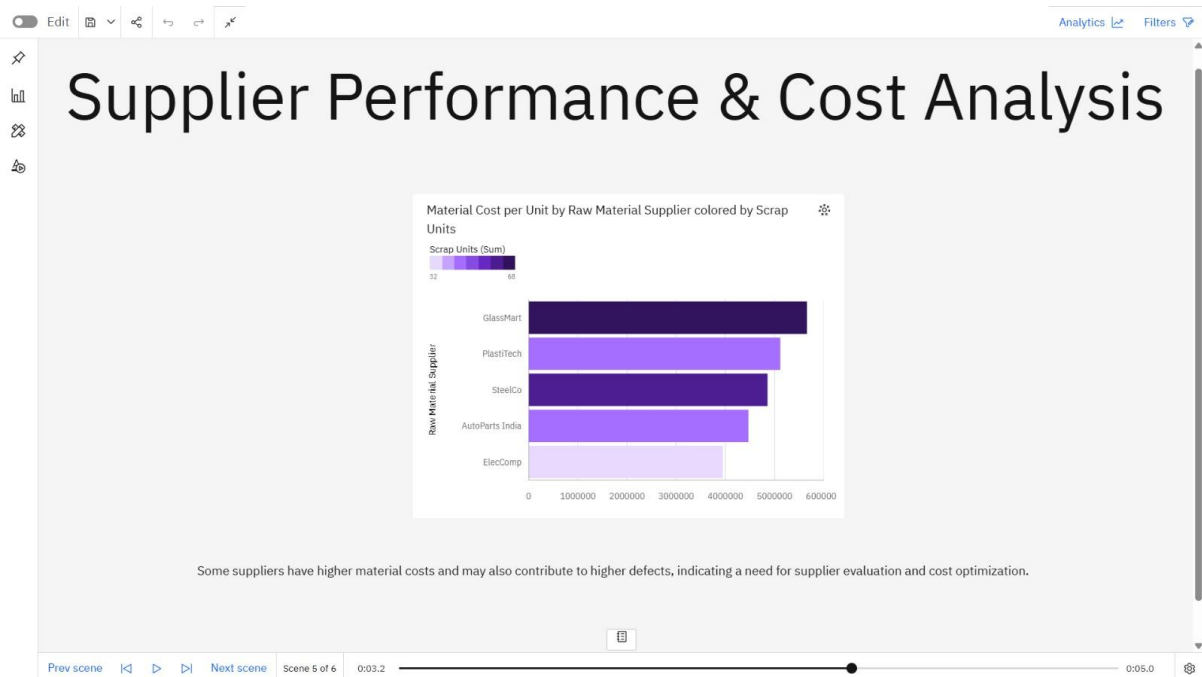
Quality analytics story highlighting defect trends, defect distribution patterns, and key areas requiring quality improvement initiatives.

9. Executive Story – Downtime Impact Analysis



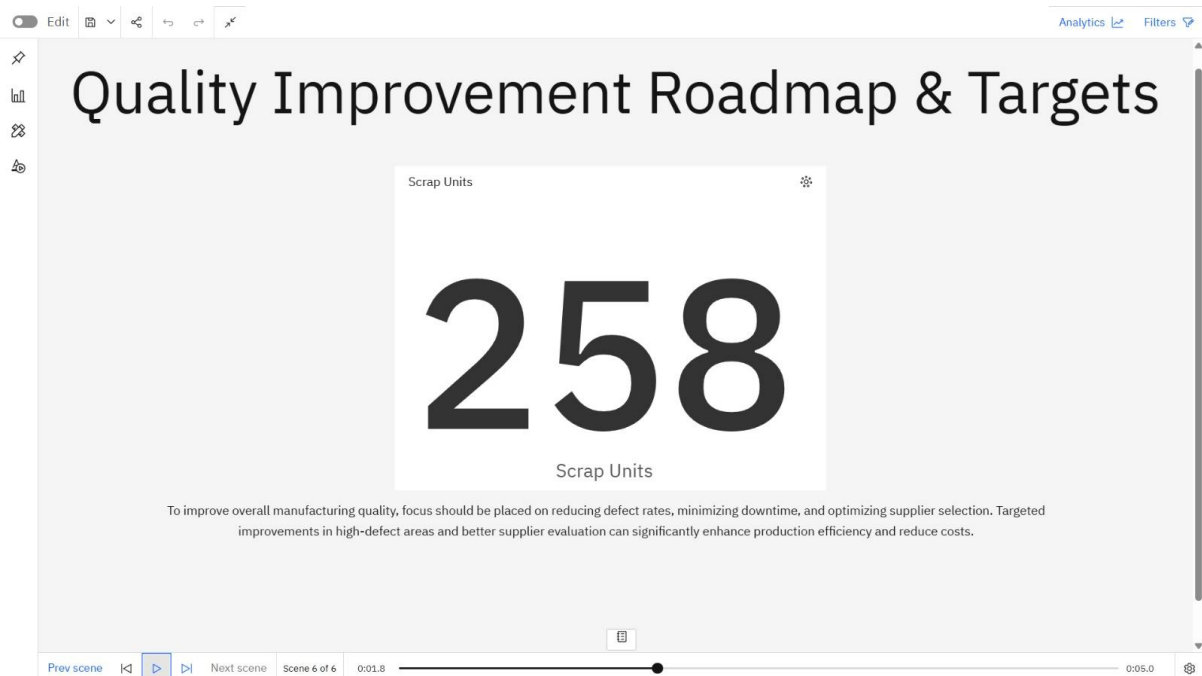
Business intelligence report illustrating the impact of downtime on production output and identifying opportunities for operational optimization

10. Executive Story – Supplier Performance Analysis



Supplier performance and cost analysis dashboard used to evaluate material costs, supplier effectiveness, and procurement optimization opportunities

11. Executive Story – Quality Improvement Roadmap



Strategic quality improvement roadmap presenting key findings, performance metrics, and recommendations for enhancing manufacturing efficiency and product quality.

Conclusion

This project demonstrated the application of IBM Cognos Analytics for manufacturing quality monitoring, operational reporting, dashboard development, and business intelligence. Through production quality analysis, downtime monitoring, supplier evaluation, and executive storytelling, the solution provided actionable insights to support data-driven decision-making and continuous improvement.